Module 1

Introduction to Blockchain Technology: History and evolution of blockchain, Centralized vs. decentralized systems, Blockchain architecture: blocks, nodes, ledgers, Public, private, and consortium blockchains, Smart contracts basics.

Introduction to Blockchain Technology

Disadvantages of the current transaction system:

- Cash can only be used in low-amount transactions locally.
- The time between transaction and settlement can be long.
- The need for a third party for verification and execution of Transactions makes the process complex.
- Fraud, cyberattacks, and even simple mistakes add to the cost and complexity of doing business, and they expose all participants in the network to risk if a central system, such as a bank, is compromised.
- Organizations doing validation charge high process thus making the process expensive.

Introduction to Blockchain Technology

Blockchain is a revolutionary technology that functions as a shared, immutable digital ledger. The name "blockchain" comes from its structure data is organized in blocks, with each new block linked to the one before it, forming a continuous chain.

Each block contains crucial data, such as a list of transactions, a timestamp, and a unique identifier called a cryptographic hash. This hash is generated from the block's contents and the hash of the previous block, ensuring that each block is tightly connected to the one before it.

- Blockchain's linked structure makes data tampering detectable by altering hashes and breaking the chain.
- It acts as a distributed database, storing transactions across the network.
- Each transaction is verified by the majority, ensuring legitimacy.
- This decentralization prevents any single party from manipulating the data.

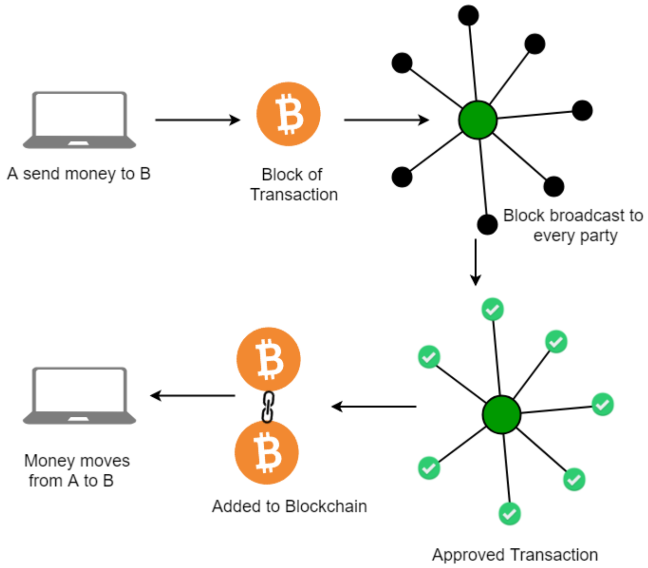
Key Characteristics:

- **Open:** Anyone can access blockchain.
- **Distributed or Decentralised:** Not under the control of any single authority.
- **Efficient:** Fast and Scalable.
- **Verifiable:** Everyone can check the validity of information because each node maintains a copy of the transactions. Permanent: Once a transaction is done, it is persistent and can't be altered.

Blockchain can be defined as the Chain of Blocks that contain some specific Information. Thus, a Blockchain is a ledger, i.e., a file that constantly grows and keeps the record of all transactions permanently. This process takes place in a secure, chronological (Chronological means every transaction happens after the previous one) and immutable way. Each time when a block is completed in storing information, a new block is generated.

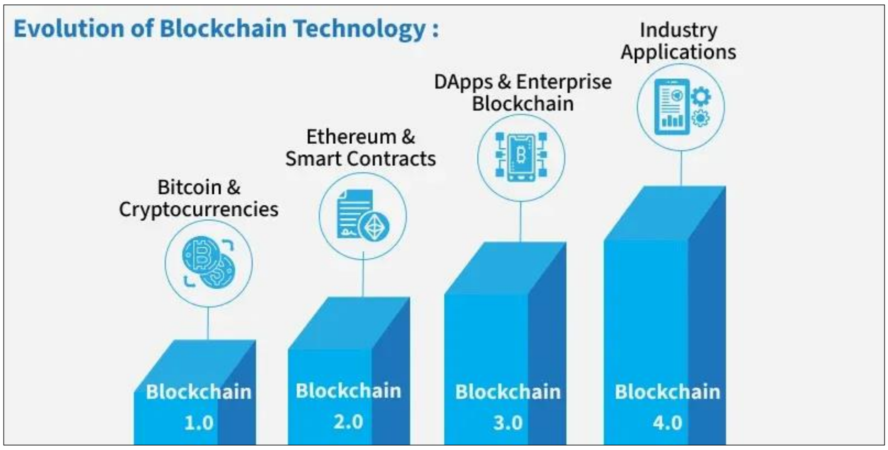
How does Blockchain Technology Work?

One of the famous uses of Blockchain is **Bitcoin**. Bitcoin is a cryptocurrency and is used to exchange digital assets online. Bitcoin uses cryptographic proof instead of third-party trust for two parties to execute transactions over the Internet. Each transaction protects through a digital signature.



History and evolution of blockchain

In 1982, David Chaum proposed a protocol with early blockchain-like features. In the 1990s, Stuart Haber and W. Scott Stornetta developed a hash-based, time-stamped chain to prevent tampering, later enhanced by Merkle trees in 1992 for efficiency.



Bitcoin was created in 2008 when an unknown entity published a white paper, using the name Satoshi Nakamoto. Bitcoin introduced a decentralized, public blockchain that solved the double-spending problem in digital money, allowing secure transactions without relying on banks or governments.

In 2015, Vitalik Buterin launched Ethereum, which expanded blockchain's capabilities beyond cryptocurrency by introducing smart contracts. This marked the shift to Blockchain 2.0, where developers could build decentralized applications (DApps) on the blockchain.

Centralized vs. decentralized systems

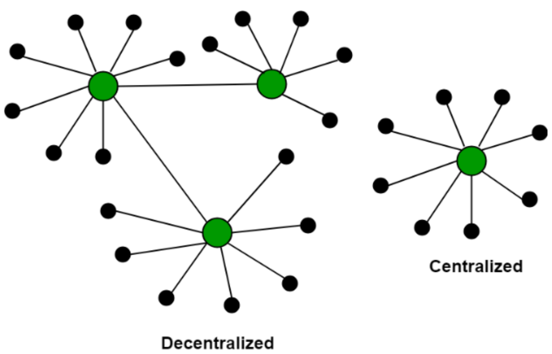
Centralized System

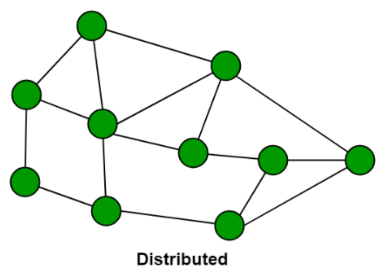
- Controlled by a single authority (e.g., a bank or company like Google).
- Data is stored in one central server.
- Faster and easier to manage.
- Less transparent and more vulnerable to single-point failure or attacks.

Decentralized System (Blockchain)

- No single authority; control is distributed across many nodes (e.g., Bitcoin network).
- Data is shared and verified by multiple participants.
- Highly transparent and secure.
- Slower and more complex due to consensus mechanisms.

There is no Central Server or System which keeps the data of the Blockchain. The data is distributed over Millions of Computers around the world which are connected to the Blockchain. This system allows the Notarization of Data as it is present on every Node and is publicly verifiable.





Blockchain Architecture:

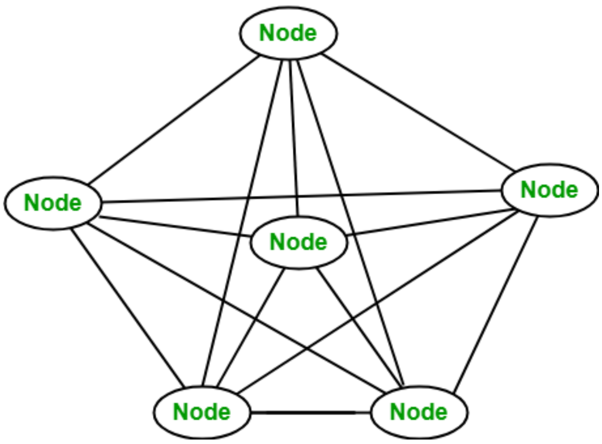
Key Components of Blockchain

- **Blockchain Blocks**

A block in a blockchain network is similar to a link in a chain. In the field of cryptocurrency, blocks are like records that store transactions like a record book, and those are encrypted into a hash tree. There are a huge number of transactions occurring every day in the world. The users need to keep track of those transactions, and they do it with the help of a block structure. The block structure of the blockchain is mentioned in the very first diagram in this article.

- **Blockchain nodes**

A node is a computer connected to the Blockchain Network. Node gets connected with Blockchain using the client. The client helps in validating and propagating transactions onto the Blockchain. When a computer connects to the Blockchain, a copy of the Blockchain data gets downloaded into the system and the node comes in sync with the latest block of data on Blockchain. The Node connected to the Blockchain which helps in the execution of a Transaction in return for an incentive is called Miners.



- **Blockchain ledgers**

Distributed Ledger Technology (DLT) is centered around an encoded and distributed database where records regarding transactions are stored. A distributed ledger is a database spread across various computers, nodes, institutions, or countries and accessible by multiple people around the globe.

Key Features:

- **Decentralized:** It is a decentralized technology where every node maintains its own copy of the ledger, and any data change triggers an independent update on each node. Even small changes are quickly reflected, and the update history is shared with all participants within seconds.
- **Immutable:** Distributed ledger uses cryptography to create a secure database in which data once stored cannot be altered or changed.

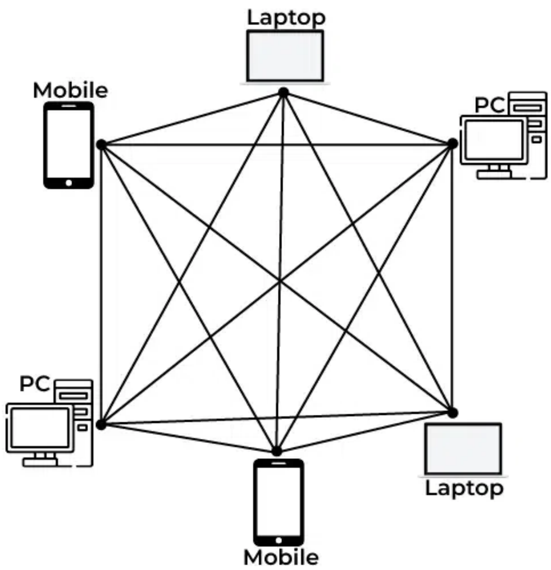
- **Append only:** Distributed ledgers are append-only in comparison to the traditional database where data can be altered.
- **Distributed:** In this technology, there is no central server or authority managing the database, which makes the system more transparent. Instead of relying on a single authoritative copy, the ledger follows specific rules for updates, ensuring decentralization. Every node verifies transactions using consensus algorithms or voting, depending on the ledger's rules. For example, Bitcoin uses the Proof of Work consensus mechanism for node participation.
- **Shared:** The distributed ledger is not associated with any single entity. It is shared among the nodes on the network where some nodes have a full copy of the ledger while some nodes have only the necessary information that is required to make them functional and efficient.
- **Smart Contracts:** Distributed ledgers can be programmed to execute smart contracts, which are self-executing contracts with the terms of the agreement between buyer and seller being directly written into lines of code. This allows for transactions to be automated, secure, and transparent.
- **Fault Tolerance:** Distributed ledgers are highly fault-tolerant because of their decentralized nature. If one node or participant fails, the data remains available on other nodes.
- **Transparency:** Distributed ledgers are transparent because every participant can see the transactions that occur on the ledger. This transparency helps in creating trust among the participants.
- **Efficiency:** The distributed nature of ledgers makes them highly efficient. Transactions can be processed and settled in a matter of seconds, making them much faster than traditional methods.
- **Security:** Distributed ledgers are highly secure because of their cryptographic nature. Every transaction is recorded with a cryptographic signature that ensures that it cannot be altered. This makes the technology highly secure and resistant to fraud.

Types of Blockchain Architecture:

1. Public Blockchain

A public blockchain is a concept where anyone is free to join and take part in the core activities of the blockchain network. Anyone can read, write, and audit the ongoing activities on a public blockchain network, which helps to achieve the self-determining, decentralized nature often authorized when blockchain is discussed. Data on a public blockchain is secure as it is not possible to modify once they are validated. The public blockchain is fully decentralized, it has access and control over the ledger, and its data is not restricted to persons, is always available and the central authority manages all the blocks in the chain. There is publicly running all operations. Due to no one handling it singly then there is no need to get permission to access the public blockchain. Anyone can set his/her node or block in the network/ chain.

After a node or a block settle in the chain of the blocks, all the blocks are connected like peer-to-peer connections. If someone tries to attack the block then it forms a copy of that data and it is accessible only by the original author of the block.



Advantages:

1. **Decentralization:** High level of decentralization, which reduces the risk of single points of failure and increases security.
2. **Transparency:** All transactions are visible to anyone, enhancing transparency and trust.
3. **Immutability:** Once data is recorded, it cannot be altered or deleted, providing a permanent record.

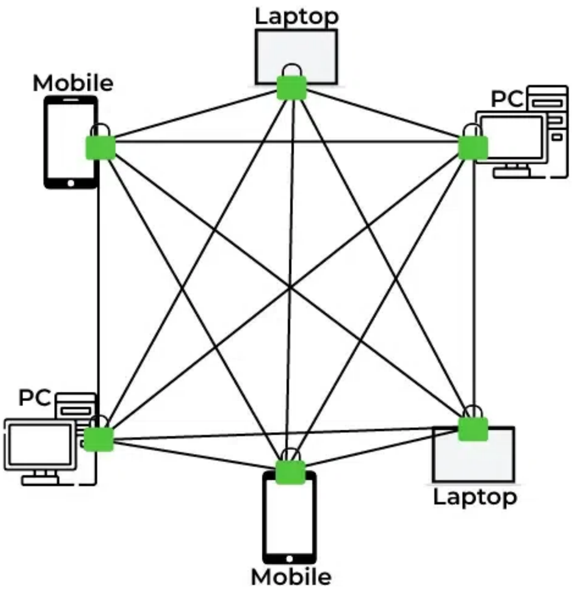
Disadvantages:

1. **Scalability Issues:** Public blockchains often face scalability problems, with limited transaction throughput and slower processing times.
2. **Energy Consumption:** Some consensus mechanisms, like Proof of Work (PoW), require significant computational power and energy.
3. **Privacy Concerns:** Public visibility of transactions may lead to privacy issues, as sensitive data can be exposed.

2. Private Blockchain

Miners need permission to access a private blockchain. It works based on permissions and controls, which limit participation in the network. Only the entities participating in a transaction will know about it and the other stakeholders not be able to access it.

It works based on permissions due to this it is also called a permission-based blockchain. Private blockchains are not like public blockchains it is managed by the entity that owns the network. A trusted person is in charge of running of the blockchain it will control who can access the private blockchain and also control the access rights of the private chain network. There may be a possibility of some restrictions while accessing the network of the private blockchain.



Advantages:

1. **Performance and Speed:** Faster transaction processing and higher throughput compared to public blockchains due to fewer nodes and reduced consensus requirements.
2. **Privacy:** Transactions and data are visible only to authorized participants, enhancing privacy.
3. **Control:** Centralized control allows for easier governance and compliance with regulations.

Disadvantages:

1. **Centralization:** Less decentralized than public blockchains, which can introduce single points of failure and reduce the security benefits.
2. **Trust:** Requires participants to trust the central authority or consortium managing the blockchain.
3. **Limited Transparency:** Reduced transparency can make it harder for external auditors to verify data.

3. Consortium Blockchain

A consortium blockchain is a concept where it is permissioned by the government and a group of organizations, not by one person like a private blockchain. Consortium blockchains are more decentralized than private blockchains, due to being more decentralized it increases the privacy and security of the blocks. Those like private blockchains connected with government organizations' block networks.

Consortium blockchains lie between public and private blockchains. They are designed by organizations and no one person outside of the organizations can gain access. In Consortium blockchains all companies in between organizations collaborate equally. They do not give access from outside of the organizations/ consortium network.

Advantages:

- **Efficiency:** Better performance and efficiency than public blockchains due to fewer nodes and optimized consensus mechanisms.
- **Shared Control:** Governance is shared among consortium members, which can enhance trust and cooperation.
- **Privacy and Security:** Improved privacy and security compared to public blockchains, as access is restricted.

Disadvantages:

- **Complex Governance:** Decision-making can be complex due to multiple stakeholders with potentially conflicting interests.
- **Less Decentralization:** While more decentralized than private blockchains, consortium blockchains still have a limited number of participants, which can reduce some decentralization benefits.
- **Interoperability:** Challenges can arise when integrating with other blockchain networks or systems.

Smart Contracts Basics

A Smart Contract (or cryptocontract) is a computer program that directly and automatically controls the transfer of digital assets between the parties under certain conditions. A smart contract works in the same way as a traditional contract while also automatically enforcing the contract. Smart contracts are programs that execute exactly as they are set up (coded, programmed) by their creators. Just like a traditional contract is enforceable by law, smart contracts are enforceable by code.

The bitcoin network was the first to use some sort of smart contract by using them to transfer value from one person to another.

The smart contract involved employs basic conditions like checking if the amount of value to transfer is actually available in the sender account.

Later, the Ethereum platform emerged which was considered more powerful, precisely because the developers/programmers could make custom contracts in a Turing-complete language.

It is to be noted that the contracts written in the case of the bitcoin network were written in a Turing-incomplete language, restricting the potential of smart contracts implementation in the bitcoin network.

There are some common smart contract platforms like Ethereum, Solana, Polkadot, Hyperledger fabric, etc.